Per-category supplies report

AI Supply Chain Terminal

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1. What this pass did

`supplies` was one undifferentiated bucket per target. Equipment, wafers, and memory all landed in the same place, so concentration could only mean "share of the target's total purchases." The paper's chokepoints are categorical: ASML is ~100% of TSMC's **lithography**, ~40% of its total tool spend. Only the 40% was representable — so ASML read as an ordinary large supplier.

This pass:

1. Adds `supply_category` to `Edge` (schema + frontend). Values used: `lithography`, `deposition`, `etch`, `inspection`, `foundry_wafers`, `memory`, `packaging`, `power_equipment`, `power_generation`, `cooling`, `gpu_accelerators`, `ai_asics`. An edge with no category falls into a `general` sub-bucket and behaves as it did before.
2. Adds per-category HHI within the `supplies` stage: split by category, compute HHI per category, combine via `max`. Same reasoning as per-stage HHI at the edge-type level — a target is fragile if *any* single category is single-sourced. Config-gated (`enabled: true` default).
3. Assigns categories to all 77 `supplies` edges.
4. Reverts the two forced values from the cleanup pass — `TSMC → NVIDIA` back to 0.99, `SK Hynix → NVIDIA` restated as **share of NVIDIA's memory supply** (not aggregate) at 0.60. Strips the pipe-concatenated cleanup note on TSMC. Corrects the SK Hynix note (bucket no longer sums to 1.00).
5. Re-authors 23 equipment-input edges (ASML/AMAT/LRCX/KLA/TEL to fabs) to per-category readings, restoring the paper-anchored ASML lithography share (0.99 at TSMC) that the aggregate model could not represent.

6. Scopes the bucket-sum test to `SUPPLY_EDGE_TYPES - supplies`; the supplies invariant now applies **per category**, in a separate test.
7. Adds `test_gallium_mined_by_hhi_unchanged` — regression guard on paper §1A's gallium anchor, since this pass only touches supplies.

Assertions:

```
PASS  A1  no per-target, per-category supplies bucket exceeds 1.0
PASS  A2  e:tsmc-supplies-nvidia input_share == 0.99
PASS  A3  no source_note contains a pipe-concatenated amendment
PASS  A4  all 63 nodes score and tier under normalize=true
PASS  A5  all 63 nodes score and tier under normalize=false
PASS  A6  gallium mined_by_hhi unchanged (0.9704, both before and
after)
```

2. The `located_in` finding

Required by §4 of the spec. **`located_in` is NOT in `SUPPLY_EDGE_TYPES`** (`backend/app/schema/enums.py`). The scoring engine's `compute_stage_hhis` only iterates supply-edge types, so location has never fed severity via concentration.

The reason my earlier bucket-sum test caught `country_region:usa located_in = 6.00` was purely a test-scope defect — the test was iterating all edge types instead of the share-semantic ones. Scoped to `SUPPLY_EDGE_TYPES`, the `located_in` overshoots are correctly excluded. **No live scoring bug.**

3. Category assignment table

77 supplies edges, 12 categories. Paper-supported taxonomy: paper §2D anchors ASML / AMAT / LRCX / KLA / TEL as the equipment stack; the split into lithography / deposition / etch / inspection is the industry- standard reading of what each vendor sells.

Category	Count	Members (source → target, input_share)
`lithography`	5	ASML → TSMC 0.99, ASML → Samsung 0.95, ASML → Intel 0.95, ASML → SK Hynix 0.55, ASML → Micron 0.50

Category	Count	Members (source → target, input_share)
`deposition`	8	AMAT / TEL → TSMC, Samsung, SK Hynix, Micron, Intel — see §5 for values
`etch`	4	LRCX → TSMC 0.80, → Samsung 0.75, → SK Hynix 0.85, → Micron 0.85
`inspection`	4	KLA → TSMC 0.90, → Samsung 0.80, → SK Hynix 0.75, → Intel 0.85
`foundry_wafers`	8	TSMC → NVIDIA 0.99, TSMC → AMD 0.98, TSMC → Broadcom 0.95, TSMC → Amazon/Google/Meta/Microsoft 0.30 each, Samsung → NVIDIA 0.01
`memory`	4	SK Hynix → HBM 0.60, Micron → HBM 0.21, Samsung → HBM 0.19, **SK Hynix → NVIDIA 0.60**
`packaging`	2	TSMC → CoWoS 0.95, Samsung → CoWoS 0.05
`gpu_accelerators`	8	NVIDIA / AMD → hyperscalers + AI labs
`ai_asics`	3	Broadcom → Google 0.20, Broadcom → Meta 0.15, Marvell → Amazon 0.10
`power_equipment`	16	Siemens Energy / GE Vernova / Quanta → utilities + data centers
`power_generation`	11	Merchant + regulated utilities → data centers, Constellation → Microsoft
`cooling`	4	Vertiv → each data center

4. Per-category bucket sums per target — the invariant holds

53 per-category buckets in total; **0 exceed 1.0**. Selected rows:

Target	Category	Sum
TSMC	lithography	0.99
TSMC	deposition	0.85 (AMAT 0.55 + TEL 0.30)
TSMC	etch	0.80 (LRCX)
TSMC	inspection	0.90 (KLA)
Samsung	lithography	0.95
Samsung	deposition	0.95 (AMAT 0.60 + TEL 0.35)
Samsung	etch	0.75
Samsung	inspection	0.80
SK Hynix	lithography	0.55
SK Hynix	etch	0.85
NVIDIA	foundry_wafers	**1.00** (TSMC 0.99 + Samsung 0.01)
NVIDIA	memory	0.60 (SK Hynix, incomplete)
HBM	memory	**1.00** (SK Hynix 0.60 + Micron 0.21 + Samsung 0.19)
CoWoS	packaging	**1.00** (TSMC 0.95 + Samsung 0.05)

TSMC 0.99 and SK Hynix 0.60 coexist on NVIDIA without displacement, because they live in different categories. The displacement problem is gone.

5. ASML — did it return to critical?

Yes. Severity 0.204 (high) → **0.539 (critical)**.

The mechanism, walked through:

- ASML has zero inbound edges — no upstream suppliers in the graph. Its severity is entirely outbound-driven.

- Under aggregate `supplies`, ASML → TSMC was 0.40 (share of TSMC's total tool spend). Under per-category, ASML → TSMC lithography is 0.99 (near-monopoly of TSMC's litho tool spend, paper §2B).
- ASML's outbound criticality walks max-product paths downstream: `ASML → TSMC → NVIDIA → OpenAI`. Before: 0.40 × 0.94 × 0.70 = 0.263. After: 0.99 × 0.99 × 0.70 = 0.686. Raw outbound roughly 2.6× higher; ASML is the graph's outbound maximum, normalized to 1.000.
- Combined concentration (max of inbound, outbound) = 1.000. Severity = 1.000 × (1 − 0.35 substitutability) × 0.83 lead_time = 0.539.

TSMC's per-category HHI for `supplies` is now 1.0 in three of four categories (litho, etch, inspection are each single-sourced by paper / industry reading). TSMC's `supplied_by_hhi` = 1.000 (was ~0.98 aggregate). Samsung's likewise = 1.000. The paper's chokepoint story is now representable — not because a value was tuned, but because the model learned the per-category split.

6. NVIDIA — the displacement problem is resolved

```
                       supplies HHI  tier      severity
Before (aggregate)     0.886        moderate  0.094
After  (per-category)  1.000        moderate  0.106

Per-category HHIs (normalize=true):
  foundry_wafers  0.9802  (TSMC 0.99 + Samsung 0.01)
  memory          1.0000  (SK Hynix 0.60 alone; bucket incomplete)
```

TSMC at 0.99 and SK Hynix at 0.60 **coexist**. The cleanup pass's forced reduction of TSMC → 0.94 is undone; the paper's "sole leading-edge foundry" reading is restored.

Note NVIDIA's severity actually *rises* (moderate → moderate, 0.094 → 0.106). Even though inbound is now higher, cascade + severity readings are unchanged for chip designers because their outbound is small (NVIDIA sells to a few large customers, none downstream-cascade-critical). No tier change.

HBM stays at high (severity 0.178) — same as post-cleanup. Per-category doesn't fix HBM because its `memory` bucket sums to exactly 1.00 and its per-category HHI is identical to its aggregate HHI (memory is the only category).

HBM’s inbound_hhi = 0.44 = “high,” not critical. Moving HBM to critical requires either `output_share` in scoring (SK Hynix → NVIDIA memory reliance) or per-category refinement below “memory” (HBM vs DRAM). Both out of scope.

7. Paper chokepoints

Under **both** normalize modes:

Node	Tier before	Tier after	Sev before	Sev after
Gallium	critical	critical	0.480	0.480
Dysprosium	critical	critical	0.545	0.545
TSMC	critical	critical	0.469	0.469
ASML	**high**	**critical**	0.204	**0.539**
HBM	high	high	0.178	0.178
CoWoS	critical	critical	0.313	0.313

6 of 6 paper chokepoints hold their paper-implied tier, except HBM. (RF & Power Semis is not a modelled node in this graph, so absent from the table.)

8. Full 63-node before/after — tier changes

normalize=true (default): 7 tier changes, all upward

Node	Before	After	Sev Δ
ASML	high	**critical**	+0.335
KLA	none	**critical**	+0.221
Lam Research	moderate	**critical**	+0.192
Samsung Electronics	high	**critical**	+0.132
Applied Materials	moderate	**high**	+0.121
Intel	none	**moderate**	+0.068
Tokyo Electron	none	**moderate**	+0.057

Every one of these is upstream in the equipment stack. The pattern is consistent: aggregate supplies HHI *understated* concentration on suppliers of critical categories. Per-category surfaces what the paper already implied.

Unchanged tiers with severity ≥ 0.005 delta:

Node	Tier	Sev Δ
Google (Alphabet)	moderate	+0.053
Copper	high	−0.013
NVIDIA	moderate	+0.012
China	moderate	−0.010

normalize=false: 10 tier changes

Node	Before	After	Direction
ASML	high	critical	↑
Samsung	none	critical	↑↑
KLA	none	critical	↑↑
Lam Research	moderate	critical	↑
Micron	none	high	↑
Applied Materials	moderate	high	↑
SK Hynix	moderate	high	↑
Intel	none	moderate	↑
Tokyo Electron	none	moderate	↑
Constellation Energy	moderate	none	↓

Under `normalize=false`, incomplete buckets self-report — so per-category restores much more concentration to Samsung / Micron / SK Hynix (which were previously under-modelled). Constellation Energy is a boundary flip at severity 0.049 → 0.045 across the 0.050 moderate threshold.

9. Test suite results

normalize=true (default)

```
28 tests, 25 pass, 3 fail

FAIL  test_no_input_share_bucket_exceeds_one — pre-existing mineral
overshoots
       (mineral:neodymium mines 1.17, refines 1.10; mineral:dysprosium
```

```
refines
       1.01). Country + facility rows layered in the same bucket;
structural
       modelling issue on the mineral edge types. Not touched by this
pass.
FAIL  test_every_paper_chokepoint_is_critical — HBM only (ASML now
passes)
FAIL  test_no_stage_bucket_sums_below_0_80 — 34 buckets, down from 38
in the
       cleanup pass. Share-completeness backlog; deliberately the
deliverable
       of the share-completeness test.
```

normalize=false

Same three failures. New per-category test passes under both modes.

Improvement: `test_paper_chokepoints` previously listed ASML and HBM; now lists HBM only. Per-category fixes the ASML half of that failure by restoring the paper's reading directly, without tuning.

10. Files changed

```
backend/app/schema/edge.py                +8 lines  (supply_category
field)
backend/app/schema/node.py                +5 lines
(supplies_per_category_hhi cache)
backend/app/scoring/config.py             +23 lines (per_category flag
/ combine)
backend/app/scoring/engine.py             +36 lines
(compute_supplies_per_category
                                                     + integration in
refresh_all_derived)
backend/tests/test_graph_integrity.py     +45 lines (scoped bucket-sum
test,
                                                     per-category
test, gallium
                                                     regression guard)
backend/tests/fixtures/scoring.yaml       resync
backend/tests/fixtures/ai/edges.json      resync
```

`config/scoring.yaml`	`+11 lines (supplies.per_category block)`
`data/ai/edges.json`	`168 edges unchanged in count; 77 supplies edges get supply_category; 23 equipment values re-authored per-category; TSMC → NVIDIA restored 0.94 → 0.99; SK Hynix → NVIDIA 0.05 → 0.60 (memory); pipe-concat note on TSMC stripped; SK Hynix source_note corrected`
`frontend/src/types.ts`	`+1 field (supply_category on Edge)`

Cascade, outbound criticality, thresholds, `normalize` default, narration, all other frontend work — all untouched, per spec.

11. What did NOT get fixed by this pass

Written down explicitly so it doesn't get counted as a next-task premise:

- **HBM tier**. The `memory` bucket sums to exactly 1.00 with three suppliers; per-category HHI ≡ aggregate HHI for HBM. Moving HBM to critical requires `output_share` in scoring (SK Hynix → NVIDIA memory- dependence) or a sub-split of `memory` into `hbm` vs `dram`. Both are out of scope here.
- **`located_in` / mineral bucket overshoots.** Not scored today. The categorical (set-membership) and country/facility-layered issues are real modelling questions but out of scope for a supplies-only pass.
- **Cascade + outbound still read `input_share`.** The spec pinned this as out of scope. Moving them to `output_share` is a separate task.
- **Equipment vendors have no inbound.** ASML now scores 1.000 outbound and 0.000 inbound — its severity is entirely from downstream

leverage. Adding upstream (rare-earth magnets, precision optics, workforce) is a data-completeness backlog item, not a modelling defect.